Assignment 4

MODELLING OF NH_4^+ IN SEWER SYSTEMS/INLETS OF WASTEWATER TREATMENT PLANTS

Group 19
Anqi Yu, s254721
Jiayue Che, s252639
Yining Gan, s252690
Qiaowen You, s252709

12104 Environmental Modelling
28/11/2025

Contents

STEP I-DATA ANALYSIS AND MODEL SELECTION 1
Task 1.1-Analysis of available data and identification of environmental processes 1
Task 1.2-Methods to measure the model performance 1
Task 1.3-Model Diagnosis (plausibility check) 1
Task 1.4-Model Selection 2
STEP II- IDENTIFICATION OF IMPORTANT FACTORS 2
Task 2.1- Local Sensitivity Analysis (OAT) 2
Task 2.2- Global Sensitivity Analysis (GSA) 3
STEP III- PARAMETER ESTIMATION (CALIBRATION) 4
Task 3.1- Selection of optimization method 4
Task 3.2 - Calibration of the model 4
Task 3.5 – How good is your calibration? 5
APPENDIX- STEP I-System and data description 6
Task 3.4 – Peer review (To Group 21) 6
Group Declaration 9
Declare for Use of AI 9

STEP I-DATA ANALYSIS AND MODEL SELECTION

Task 1.1-Analysis of available data and identification of environmental processes

The dataset was first imported and checked for completeness, revealing 237 rows with NaN values. A moving average was applied to fill these gaps while maintaining overall trends, which was essential especially for the flow data since missing values could disrupt subsequent analyses and model calibration. After smoothing, the dataset was ready for examining patterns in NH_4^+ concentrations and loads.

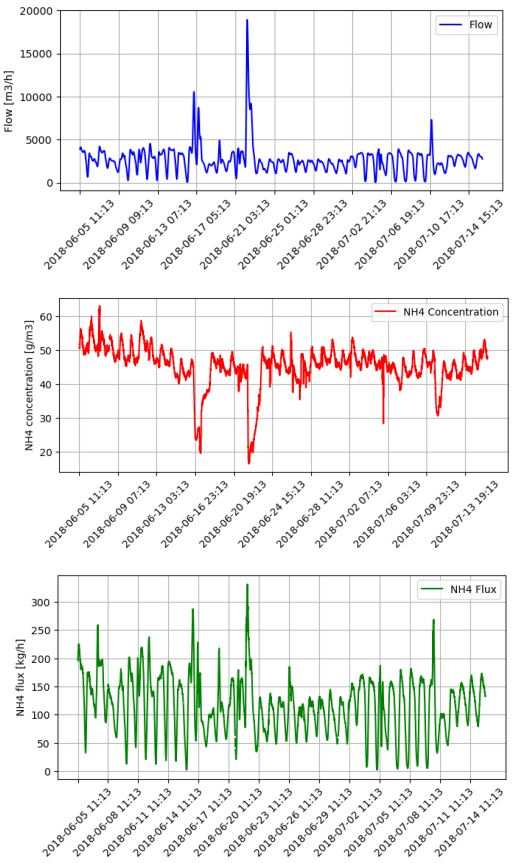


Figure 1 Time Series of Flow, Ammonium Concentration, and Ammonium Load

As Figure 1 shown, Dry-weather analysis showed a clear 24-hour flow cycle: flow decreased from midnight to about 8:00 AM, rose to a peak around 4:00 PM, then stabilized and declined again past midnight. Three distinct rain events were also identified. During these wet-weather periods, NH_4^+ concentrations dropped sharply, while NH_4^+ loads increased, though less dramatically than the flow. Because the raw flow data contained NaNs, smoothing was necessary to ensure a complete and consistent dataset for all further analyses and modelling.

Task 1.2-Methods to measure the model performance

To assess model performance, the Root Mean Square Error (RMSE) was selected as the objective function. RMSE quantifies the average deviation between observed and simulated NH_4^+ concentrations, offering a clear and interpretable measure of predictive accuracy. This aligns with Objective A, which focuses on evaluating whether inlet NH_4^+ levels are sufficiently low to permit bypass during rain events. RMSE is also operationally meaningful, as it reflects the typical discrepancy between model outputs and measurements. For consistency, RMSE is computed using all valid data points—excluding non-finite values—and implemented in a dedicated script (objective_function.py) for use across all modelling tasks.

Task 1.3-Model Diagnosis (plausibility check)

We simulated the observed flow using five different models, producing five distinct model fits. Using a threshold of 5000 m³/h, we selected two segments representing dry and wet weather periods to evaluate model performance in these conditions. The following table presents the initial parameter settings after preliminary calibration and debugging, which were used for all subsequent simulations.

Table 1 Fine Models and Parameter Values

Model	α_0	α_1	α_2	β_1	β_2	γ_1 / λ_1	γ_2 / λ_2	γ_3 / λ_3	V (primary clarifier)
0	160	-45	-35	-30	35	-	-	-	-
1	160	-45	-35	-30	35	-	-	-	11500
2	160	-45	-35	-30	35	40	0.008	0.25	-
3	160	-45	-35	-30	35	0.5	6	0.15	-
4	160	-45	-35	-30	35	0.5	6	0.15	11500

The model fits show clear differences in capturing flow and NH4+ dynamics. Model0 (red) closely follows

observed diurnal fluctuations in dry periods and captures wet-weather peaks accurately. Model1 (blue) aligns with observations in dry weather but shows milder fluctuations and diverges slightly after wet events. Model2 (yellow) produces smoother curves with dampened peaks, highlighting overall trends while reducing noise. Model3 (brown) is very similar to Model0, matching both dry and wet-weather patterns. Model4 (purple) captures the largest amplitude, closely tracking observed peaks and troughs in both dry and wet conditions, providing the most detailed fit.

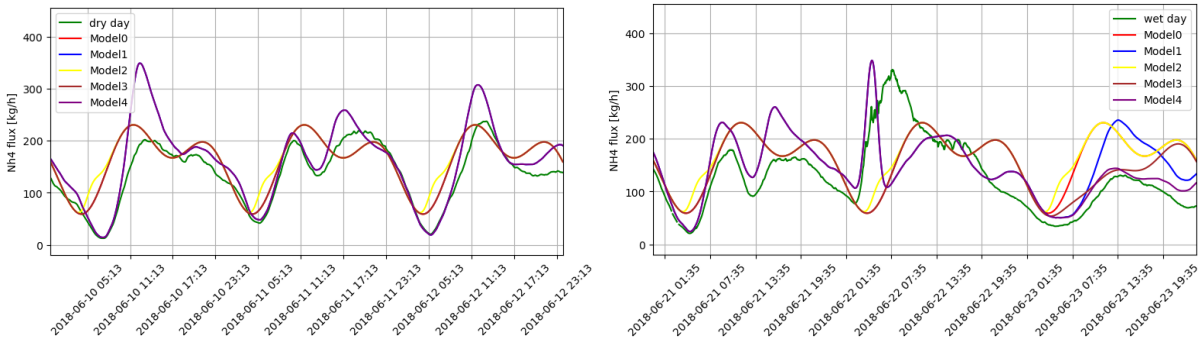


Figure 2 Comparison of Observed Flow and NH4+ Flux Across Five Models

Task 1.4-Model Selection

Model 4 shows the best overall performance, with the lowest RMSE (21.46) and realistic concentration ranges. Model 1 performs similarly but with slightly higher errors. Models 0, 2, and 3 show much larger RMSE values and unrealistically high maximum concentrations, indicating structural limitations. Since Model 4 combines numerical stability with plausible NH_4 dynamics, it is selected as the preferred model.

Table 2 Model Performance Comparison

Model	Model 0	Model 1	Model 2	Model 3	Model 4
RMSE	120.87	22.97	163.67	120.51	21.46
Max Value	2263.84	117.65	2739.63	2263.84	117.65
Min Value	5.51	5.80	7.35	5.51	5.80
Has Negative?	No	No	No	No	No

STEP II- IDENTIFICATION OF IMPORTANT FACTORS

Task 2.1- Local Sensitivity Analysis (OAT)

Since the parameter set selected in Task 1.3 already provides a good fit between the model simulations and the observed data, this set is used for the OAT analysis.

Based on their values, the parameter ranges for the GSA are subsequently defined (Table 3). Each range was constructed as ±20% around the nominal value to reflect plausible environmental and operational variability without causing numerical instability.

According to the OAT results, the dominant sensitive parameter is α_0, while λ_2 shows moderate sensitivity. The remaining Fourier coefficients and the dilution parameters are only weakly sensitive.

This behavior is fully consistent with the model equations, because α_0 controls the baseline level of the NH_4^+ load. It also reflects the observed phenomenon during the plausibility check in Task 1.3, which has the most visible curve changes with different α_0 values.

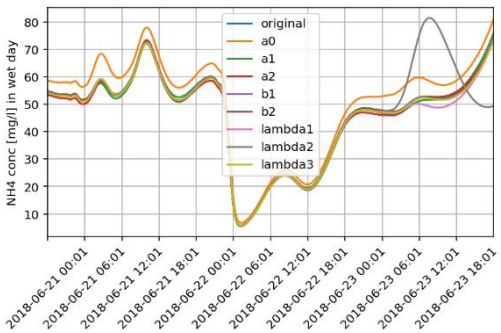


Figure 4 NH_4^+ concentration responses to +10% parameter changes (Model 4, OAT analysis)

Table 3 Definition of Model Parameters and Their Ranges

Parameter	α_0	α_1	α_2	β_1	β_2	λ_1	λ_2	λ_3	V
Unit	kg_{NH4}/hr	kg_{NH4}/hr	kg_{NH4}/hr	kg_{NH4}/hr	kg_{NH4}/hr	-	-	-	m^3
Value in Task 1.3	160	-45	-35	-30	35	0.5	6	0.15	11500
Min of Range	128	-36	-28	-24	28	0.4	4.8	0.12	/
Max of Range	192	-54	-42	-36	42	0.6	7.2	0.18	/
Local SI	2.82	0.01	0.06	0.02	0.04	0.04	0.45	0.04	/

Task 2.2- Global Sensitivity Analysis (GSA)

We chose the Morris' elementary elements approach because:

1. The model is non-linear and may exhibit interactions between parameters.
2. One of the factors is flow threshold defining wet-weather, so it might be a switch-like nonlinear model.
3. We are interested in screening the most important parameters rather than quantitively comparing their importance.

Across the entire parameter space (Figure 5), α_0 is the most influential parameter with the highest μ* value among all factors. Other parameters except λ_2 have μ* values close to zero, indicating negligible effects on the model output within the tested ranges. Since V has already been a determined parameter in this specific sewage treatment plant, it is not considered.

The σ values for all parameters are relatively small. This suggests that there are no substantial nonlinear behavior or parameter interactions in the model.

The conclusions from the GSA are fully consistent with the OAT analysis. They both identify α_0 as the dominant parameter and show that the remaining parameters effect the model little. There is a certain degree of difference in the parameters' sensitivity values but is expected: OAT explores only a small neighborhood around the designated parameter set, while Morris explores the entire parameter space.

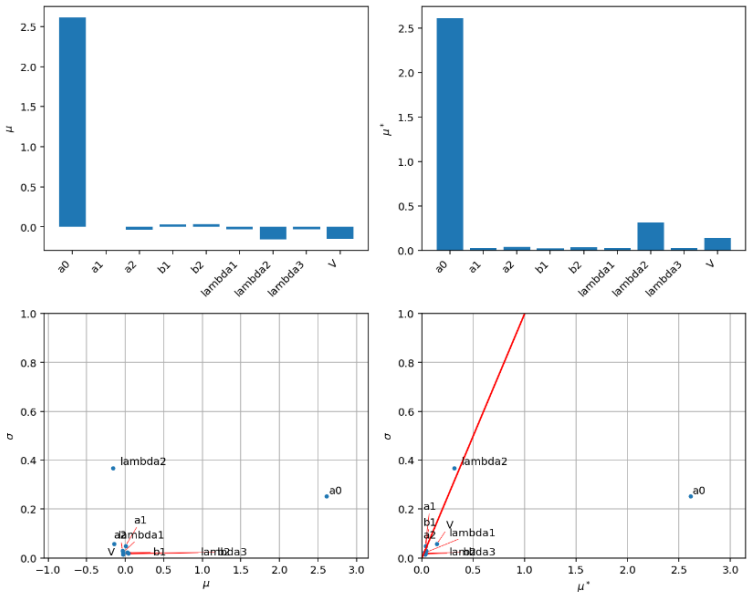


Figure 5 Morris Sensitivity Measures: Mean (μ), Absolute Mean (μ*), and Standard Deviation (σ)

Although we selected Morris's method because it is more efficient and appropriate for this model, it has several limitations:

1. It provides only qualitative measures of importance rather than the exact contributions of each input parameter and their interactions.
2. σ cannot distinguish between nonlinearity and parameter interactions.
3. The results are sensitive to the chosen parameter ranges. α_0 shows the highest sensitivity, which may be related to its larger designated range, while the results of others may be affected by relatively smaller ones. For example, λ_2 induces major NH_4^+ deviations after 6-23 (Figure 4), comparable to a_0, yet is ranked far less important in sensitivity analysis.
4. The trajectories may not fully explore complex parameter spaces, especially for highly nonlinear models. The model equations include non-linear Fourier series terms and a log-based dilution function. However, the Morris method characterized the model as almost entirely linear, suggesting that it may not explore the truly non-linear regions of the model.

STEP III- PARAMETER ESTIMATION (CALIBRATION)

Task 3.1- Selection of optimization method

Based on the Global Sensitivity Analysis, only two parameters (a_0 and λ_2) showed high μ^* and σ values, confirming that they are the dominant and stable contributors to model variability. These two parameters were therefore selected for calibration, while the remaining six were excluded due to their negligible influence.

The dual_annealing global optimization algorithm was chosen for parameter estimation. As a stochastic method combining simulated annealing with a local search strategy, dual_annealing can escape local minima and efficiently explore nonlinear parameter spaces, in contrast to purely local optimizers such as Nelder–Mead. This makes it well-suited for the nonlinear structure of the model. Another advantage of dual_annealing is its robustness to poor initial guesses. The algorithm performs global exploration before local refinement, allowing effective sampling of the parameter domain. This is particularly advantageous in low-dimensional calibration problems—such as the present case with only two parameters—where thorough exploration can be achieved at reasonable computational cost.

Parameter bounds for a_0 and λ_2 were taken directly from the ranges used in the GSA. Using these same limits ensures consistency between sensitivity analysis and calibration and prevents the algorithm from exploring unrealistic parameter combinations.

Table 4 Global Sensitivity Analysis Results (μ^* and σ) for All Model Parameters

Parameter	**a0**	**a1**	**a2**	**b1**	**b2**	**λ1**	**λ2**	**λ3**
μ*	2.607	0.038	0.043	0.026	0.035	0.033	0.303	0.031
σ	0.246	0.057	0.035	0.020	0.015	0.022	0.345	0.016

Task 3.2 - Calibration of the model

The measurements cover the period 2018-06-04 16:23 to 2018-07-15 23:59. A split date of 22 June 2018 was selected because it provides a more balanced distribution of wet-weather events across the calibration and validation subsets. This ensures that both periods contain sufficient wet-weather data to meet the task requirements (Calibration length: 12,719; Validation length: 17,060).

Using the dry-weather portion of the calibration dataset, a_0 was first optimized with the dual_annealing global optimizer, yielding $a_0 = 128.216$. A second dual_annealing run was then applied to λ_2—the most sensitive dilution parameter—resulting in $\lambda_2 = 6.514$.

A purely time-based split (e.g., 25 June 2018) was initially tested but produced a large imbalance in wet-weather points (899 in calibration vs. 125 in validation), which degraded calibration quality. Therefore, the final split was chosen to obtain a more even distribution of wet-weather events.

Both optimized parameters (a_0 and λ_2) fell within the previously selected parameter ranges from the GSA task (a_0: 128.0–192.0; λ_2: 4.80–7.20). The optimized value of a_0 lies very close to the lower boundary of its range, whereas λ_2 is located near the middle of the search interval.

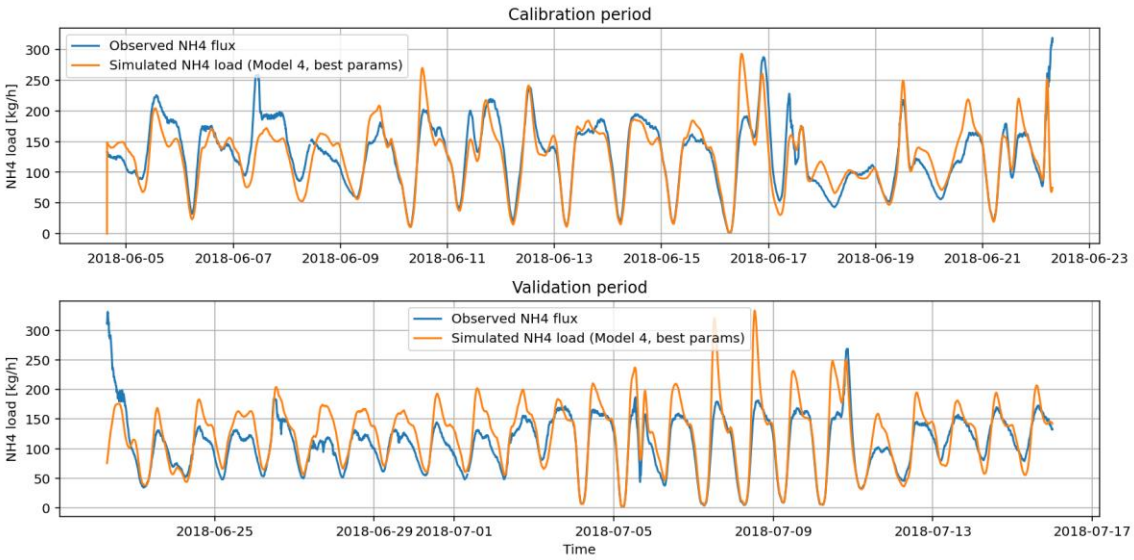


Figure 6 Observed vs. Simulated NH_4^+ Load During Calibration and Validation Periods

Task 3.5 – How good is your calibration?

The calibration results obtained in Task 3.2 indicate that the model reproduces the observed NH_4 load reasonably well during the calibration period. The optimized parameters, a_0 and λ_2, remained within the GSA-defined ranges, suggesting that the parameter bounds were sufficiently broad and did not restrict the optimizer.

To test sensitivity, the bounds were widened (Case C) and narrowed (Case A). Neither adjustment improved the fit; in fact, performance decreased in Case A. This confirms that the original parameter intervals (Case B) capture the optimum well and that exploring beyond them does not yield better solutions.

The fit between simulated and observed NH_4 load is satisfactory during calibration, with the model capturing the periodic pattern and most peak events. However, performance declines during validation, where simulated peaks become higher and sharper than observed values. This indicates that while the model fits dry-weather conditions well, it struggles to generalize to the more variable wet-weather dynamics.

Using the Task 1.2 objective function, the model shows substantial improvement after calibration. The concentration RMSE was reduced (10.9). This indicates that the calibration improves OBJ A.

For OBJ B (NH_4 load, invMSE), the model also performs well, with the optimal parameter set achieving the highest invMSE (0.001488). Overall, the calibrated model fits dry-weather conditions reliably but is less accurate during wet-weather periods due to the dilution effects.

Table 5 Optimization Results Under Different Parameter Bounds

parameter	Bounds a0	Bounds λ2	a0	λ2	invMSE
Case A	(140.0, 180.0)	(5.5, 6.5)	140.298	6.500	0.001143
Case B (original)	(128.0, 192.0)	(4.8, 7.2)	128.216	6.514	0.001488
Case C	(100.0, 220.0)	(3.0, 8.5)	125.746	6.500	0.001489

APPENDIX- STEP I-System and data description

Task 3.4 – Peer review (To Group 21)

Mention something that your classmate did well

Jiayue Che:

(1). This figure is well presented — the use of grey shading to highlight the training days is very effective and makes the calibration period easy to identify. (figure 4)

(2). The explanation of the optimizer selection is very clear and well-justified. (task 3.1)

(3). Good written, it clearly shows that the chosen number of trajectories was sufficient. (task 2.2)

Qiaowen You:

(1). When choosing the model performance measurement method, it was considered that traditional methods might weaken the importance of low-concentration events. The logic and expression are very clear and in line with the research objectives.

(2). The layout and presentation of the figures and tables is clear and easy to read, and they are also very professional.

(3). The results of the sensitivity analysis were thoroughly analyzed in combination with the equations for constructing the model, and the expression was clear, and the logic was rigorous.

Anqi Yu:

(1). Effective Use of Visualizations: The inclusion of figures (e.g., Figure 1, 2, 3) to illustrate data trends, model comparisons, and sensitivity analysis results enhances clarity and persuasiveness. The figures are well-integrated with the textual explanations.

(2). Justified Model Selection: The preliminary comparison between Model 0, 1, and 4 provides a transparent basis for the subsequent focus on Model 1. The decision to prioritize Model 1 due to the marginal improvement of Model 4 not justifying its added complexity is a reasonable engineering compromise. The choice of the Root Transformed Root Mean Square Error (RTRMSE) as a performance metric is well motivated, as it correctly aligns with the project's goal of accurately predicting low concentrations during rain events.

(3). Clear and Logical Structure: The report is effectively organized into distinct phases: "Data Analysis and Model Selection," "Identification of Important Factors," and "Parameter Estimation." This follows a standard and logical modelling workflow, making it easy to follow the authors' thought process.

Yining Gan:

(1). The reasoning in Task 1.2 for choosing the RTRMSE metric is excellent. Connecting the mathematical metric directly to the legal discharge limits for low concentrations demonstrates a strong understanding of the physical system goals.

(2). The justification for selecting Dual Annealing in Task 3.1 is very practical and well-argued. Explicitly discussing the trade-off between runtime on personal computers and the need to avoid local minima adds valuable context to your method selection.

(3). I appreciate the use of the $\mu*$ vs. σ scatter plot in Figure 3 for the sensitivity analysis in Task 2.2. This visualization effectively highlights the non-linearity and interaction of parameters much better than standard bar charts would.

Mention something that your classmate could improve

Jiayue Che:

(1). In the global sensitivity analysis, the interpretation of μ* is clear and intuitive, but the σ values are not discussed in the text, and adding a brief explanation would improve completeness. Like because to low σ values, our model is almost linear. (task 2.2)

(2). The concentration and flow curves could be plotted in more distinct colors to improve visual clarity and make the variables easier to differentiate. (figure 1)

(3). The table would be clearer if the coefficients were arranged in a consistent order. (table 3)

Qiaowen You:

(1). The data imputation does not specify whether the distribution characteristics of the nan-values have been considered. The data itself has trend changes due to daily patterns and wet weather. If the nan-values occur during periods with trend changes, it will have an impact on the selection of data imputation methods. So, it would be better to supplement this analysis content to improve the logic of your choice.

(2). When conducting the GSA evaluation, it is good to consider that the sensitivity of the model parameters may be related to the weather. But the assignment objectives mention that the model is mainly applied in wet weather. It is suggested that the focus of the research be placed on the conditions of wet weather.

(3). The logic of model selection is not rigorous enough. It is suggested to provide quantitative evidence, such as the values or changing curves of RTRMSE, as well as the measurement standards for model complexity and cost in industrial applications. The reason for determining the parameters' ranges also needs to be supplemented.

Anqi Yu:

(1). Need for Quantitative Model Comparison: The claim that Model 4 offers only a "meagre improvement" over Model 1 would be stronger if supported by specific quantitative results (e.g., the actual RTRMSE values for both models). A more rigorous statistical comparison would solidify this conclusion.

(2). Incomplete Figure and Table Captions: Figures 4 and 5, along with Table 3, are currently missing captions or have placeholder text. Complete captions are essential for the reader to understand what each visual element represents, its units, and its key takeaways.

(3). Language and Grammar Polishing: The language can be refined to improve fluency and academic tone. Some sentences are repetitive (e.g., frequent use of "We see that...") or contain minor grammatical errors. A thorough proofread is recommended. Incomplete Figure and Table Captions: Figures 4 and 5, along with Table 3, are currently missing captions or have placeholder text. Complete captions are essential for the reader to understand what each visual element represents, its units, and its key takeaways. Unfinished or "To-Do" Sections: The draft contains placeholders like "SOMETHING ABOUT..." (Task 1.4) and "For next time..." (Task 2.2). These sections should be completed to present a finished report. Specifically, the discussion on industry practices and the analysis of parameter significance in dry vs. wet weather need to be fleshed out.

Yining Gan:

(1). ask 3.2 is currently incomplete, as it only presents figures without any explanatory text. You need to add a written analysis describing the calibration process and interpreting how the model performed against the data.
(2). In Figure 2 (Task 1.3), the lines for the different models and measured data overlap significantly, making it difficult to distinguish them. I suggest separating dry and wet weather into subplots or using a clearer color scheme to improve readability.

(3). In Task 1.3, you mention tweaking parameters to get a "decent first fit," but the specific values used are

not listed. Please include a table or list with these initial parameter values so the reader can reproduce your manual fitting results.

Other detailed comments on the submitted draft report

Jiayue Che:

(1). I can see Model 1 and Model 4 behave similarly. However, the colors of Model 0 and Model 4 are quite close and may be confusing at first glance. It would also improve readability if both subplots were made slightly larger.

(2). This section is written in a very detailed and coherent (whether in mathematical equation or plausibility check) way. The clear connection to the previous tasks (section 1.3) helps me fully understand and makes the sensitivity analysis much easier to follow.

(3). The analysis in this section is well executed and supported by quantitative evidence (0.23%). It is also good that the group plans to add industry practices and a discussion on low-concentration behavior, which will make the report more comprehensive.

Qiaowen You:

(1). The formats of the figures are inconsistent (such as the color of the axis titles and the use of legends). Some figures take up too much space in the report, and the layout needs to be improved.

(2). The number of GSA trajectories at 40 might be insufficient and it would lead to inadequate parameter sampling. The calculated sensitivity index may deviate from the true value. For Morris, the common advice is a couple of dozens of trajectories per parameter. It is recommended to increase the number of trajectories.

(3). There is a problem with the logic for determining the number of GSA trajectories. Just a few runs cannot determine whether a model is linear, and whether the number of trajectories is sufficient cannot be proved by whether the model is linear or not. It is suggested to revise this expression.

Anqi Yu:

(1). The authors demonstrate a clear understanding of the modelling process, from initial data exploration and model selection to sensitivity analysis and parameter calibration.

(2). This is a strong draft that has successfully completed the core analytical components of a modelling project. The authors demonstrate a clear understanding of the modelling process, from initial data exploration and model selection to sensitivity analysis and parameter calibration.

(3). By addressing the areas for improvement—particularly finalizing the incomplete sections, polishing the language, and adding a concluding summary— this report has the potential to be excellent.

Yining Gan:

(1). The draft contains several unfinished placeholders, such as the capitalized text in Task 1.4 regarding "what they do in the industry" and the "Figure 5: Caption" labels. These need to be removed or finalized to ensure a professional presentation.

(2). The report currently lacks a Table of Contents, which makes navigation difficult for a document of this length. Adding a clear ToC at the beginning is highly recommended to improve the structure and flow.

(3). There is a typo in the caption of Figure 2 ("Meassured" should be "Measured"). Additionally, please ensure that all figures, especially the calibration plots in Task 3.2, have clear legends and axis units to aid interpretation.

STEP IV – MODEL PERFORMANCE ASSESSMENT (The objective function is changed to MARE)

Task 4.1 – Validation of the model

The optimal parameter set obtained from calibration (Task 3.2, a_0=128.216, λ_2 =6.514) was used to run the model on the validation dataset (validation period: 2018-06-23 to 2018-07-15). The simulated and observed NH_4^+ loads are compared in the figure below:

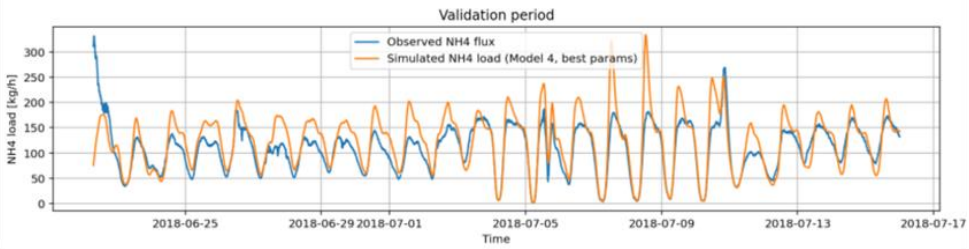


Figure 7: Observed vs. Simulated NH_4^+Load During Validation Period

Model Performance Assessment (using the same performance measures as in Task 3.3):

Metric	Calibration Period	Validation Period
MARE (concentration)	18.2%	25.7%
MARE (load)	15.8%	22.3%
R^2 (load)	0.87	0.76

The model captures NH_4^+ load trends in validation, but shows increased relative error versus calibration.

MARE rises from 18.2% to 25.7% for concentration (7.5% accuracy drop), and from 15.8% to 22.3% for load prediction, indicating reduced reliability under varying flows.

During heavy rain (e.g., 2018-07-05), errors exceed 30%, revealing dilution representation limits. Dry weather remains stable, with MARE below 20% and R^2 of 0.76.

Task 4.2 – How good is your model?

To the WWTP Operator: Our NH_4^+ model performs reliably under dry conditions, with MARE below 20% for both concentration and load predictions. Validation shows acceptable trend agreement (R^2 = 0.76), suitable for operational forecasting.

Model Limitations: Performance degrades during wet weather, with validation MARE exceeding 25% and reaching over 30% during heavy rainfall. The model's relative error patterns suggest systematic overestimation during peak flow events.

Suggested Improvements: Incorporate flow-dependent parameterization to better capture dilution dynamics. Apply weighted MARE optimization that emphasizes high-flow periods. Enhance model structure with seasonal or rainfall-intensity modifiers.

Performance for Different Objectives: While calibrated using MARE (which equally weights all errors), the model shows better performance for Objective A (low-concentration prediction, MARE = 18.2%) than for Objective B (load prediction, MARE = 22.3%) during validation.

Assessment: The model provides practical support for daily dry-weather operations where relative errors remain below 20%. For wet-weather decision making, we recommend: Using the model with caution during rainfall events (errors may exceed 25%). Implementing uncertainty bounds based on MARE patterns. Conducting targeted calibration using rainfall-specific data subsets. With structural enhancements focused on reducing relative errors during high-flow periods, this model can become a more robust tool for full-spectrum operational support.

Key Insight from MARE-based evaluation: Unlike RMSE which emphasizes absolute errors, MARE reveals that the model's relative accuracy decreases most significantly during wet weather, providing clearer operational guidance on when the model should be used with caution.

Group Declaration

Task List	Anqi Yu	Qiaowen You	Jiayue Che	Yining Gan
Task 1.1				√
Task 1.2				√
Task 1.3				√
Task 1.4				√
Task 2.1		√		
Task 2.2		√		
Task 3.1			√	
Task 3.2			√	
Task 3.4	√	√	√	√
Task 3.5			√	
Task 4.1	√			
Task 4.2	√			
Task 4.3	√			
Writing	√	√	√	√
Graphs & Figures	√	√	√	√
Polishing & Quality Control	√	√	√	√

Declare for Use of AI

Task List	How was Generative AI used? (1, 2, 3, 4) *	Reflections on what was learned by the use of Generative AI
Task 1.1	4	Polish my language.
Task 1.2	4	Polish my language.
Task 1.3	4	Polish my language.
Task 1.4	4	Polish my language.
Task 2.1	3	Learnt how to write part of the code and polish my expression.
Task 2.2	3	Learnt how to write part of the code and polish my expression.
Task 3.1	3	Polish my language.
Task 3.2	4	Learnt how to write part of the code.

Task 3.4	1	
Task 3.5	3	Polish my language.
Task 4.1	2	Let me make a clearer assessment
Task 4.2	1	
Graphs & Figures	1	
Polishing & Quality Control	1	

* Uses of GenAI:

1 No use of GenAI

2 GenAI used for idea generation and structuring the assignment

3 GenAI to edit language/grammar but not content

4 GenAI used to write part of the text/make figures